

A Haptic Device for Tennis Simulation: Dual-Flywheel System for Rendering Virtual Impact

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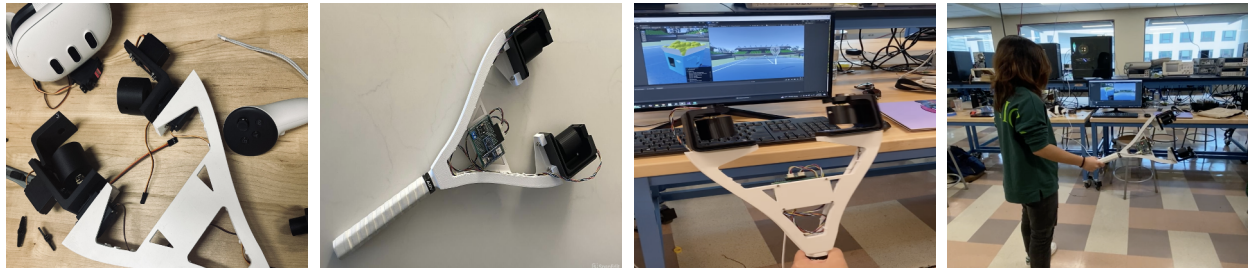


Figure 1: Overview of the haptic racket design and functionality, showing its components, assembly, virtual environment integration, and use in testing. (left to right)

This paper introduces a haptic feedback device for tennis simulation, designed to provide realistic physical sensations of ball impact through interaction with a virtual racket model. The system integrates with a Unity-based virtual setup where users are visually immersed in a virtual tennis court, complete with a racket, ball machine, and tennis balls. This system builds on immersive haptic technologies, such as those exemplified by the Nintendo Wii, which introduced real-time tactile feedback for interactive gaming. Unlike previous systems, this device incorporates a mechanical interface that replicates the sensation of a tennis ball striking the racket. As the user plays, the ball's collision with the racket is simulated and mapped to an impedance-type kinesthetic response. The system generates real-time physical feedback, enabling users to experience the sensation of ball impact, enhancing the realism of virtual tennis gameplay.

The haptic racket is designed to enable users to distinguish between impact forces at high and low accelerations, providing clear feedback during virtual gameplay. The system was developed with the goal of minimizing the delay between virtual collisions and the corresponding haptic effects; however, a slight delay remains noticeable. While the racket aims to replicate the feel of a real tennis racket, its resemblance is more apparent at lower accelerations, where the feedback aligns more closely with typical racket behavior.

This system employs a single-function design, a paradigm ideal for specialized applications like gaming peripherals, where focused functionality enhances realism, immersion, and user experience compared to general-purpose devices.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices; Human-centered computing—Haptics; Human-centered computing—Interaction paradigms—Augmented and virtual realities;

1 INTRODUCTION

Haptic devices play a crucial role in enhancing the realism of virtual experiences by providing tactile or kinesthetic feedback that mimics real-world sensations. In sports, haptic technology offers transformative opportunities to bridge the gap between physical and virtual environments. A haptic tennis racket, for example, enables players to feel the impact of hitting a virtual tennis ball, making the experience more immersive and practical for training purposes.

Timing is a critical aspect of tennis, and a haptic racket provides

invaluable feedback to help players refine the precise moment their racket should make contact with the ball. For beginner and intermediate players, this feedback can significantly enhance their ability to develop proper stroke mechanics and improve consistency in their gameplay [7].

Earlier gaming innovations like the Wii tennis racket introduced an engaging way to simulate tennis, offering players an accessible and fun way to interact with virtual gameplay [1]. However, there are several areas where the Wii racket could be improved to better support skill development and technique refinement. The Wii racket provided basic tactile feedback in the form of vibration when making contact with the ball, but this feedback lacked the nuanced sensations of impact that vary with different swing speeds and ball interactions. Additionally, the motion tracking, while a step forward, lacked the accuracy needed for precise racket movement and timing, with occasional delays and inconsistencies. The feedback was also limited in variety, offering only vibration without simulating the full range of sensations, such as force or resistance, felt during a real hit. These limitations hindered the ability to replicate the subtle physical sensations necessary for developing proper stroke mechanics and improving overall player performance.

In contrast, a well-designed haptic tennis racket addresses these gaps by delivering realistic feedback on timing, impact dynamics, and stroke technique. Beyond individual training, it supports remote coaching, rehabilitation, and enhanced player engagement, offering a versatile solution for athletes across skill levels.

The goal of this project is to create a virtual tennis court environment that provides realistic haptic feedback to the user when the racket contacts the ball. This setup allows tennis players—both beginners and advanced—to practice crucial timing and decision-making for ball impact and placement. The haptic feedback is delivered through a racket handle equipped with various instruments, such as a dual flywheel system, to simulate the sensation of ball impact. A summary of the haptic racket's design and functionality can be seen in Figure 1.

Participants will use a head-mounted display, specifically the Meta Quest 3 headset, to fully immerse themselves in the virtual tennis environment, where realistic haptic feedback enhances their training experience.

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2 BACKGROUND

A thorough literature review of existing haptic rackets and flywheel systems was conducted to inform this project. Additionally, a review of the effectiveness of haptic and visual feedback to promote learning was performed.

2.1 Haptic Rackets

There have been a number of rackets that use various haptic systems to generate impact sensations. The AirRacket used compressed air jets to provide the user with directional force feedback when striking a ball [2]. The compressed air had a limit to how much force it could output, but researchers found that with longer impact duration, the users perceived a stronger force. This idea was utilized in this project when determining how long flywheels should spin.

The Drag:on virtual reality controller also utilized airflow to generate haptic sensations. The controller changed shape as it moved, effectively changing the drag force it experienced and its moment of inertia [3]. Ideas from this paper were taken into account when designing the shape of the haptic racket.

Haptic feedback in VR game controllers like the PlayStation DualSense, Oculus Quest controllers, and Vive controllers primarily rely on DC motors with offset weights or other specialized actuators to create vibrations. This method of sensation rendering is not only energy inefficient, but also lacks the versatility to generate different sensations for different tennis strikes. [4]

2.2 Flywheel Systems

Spinshot was a single flywheel system used to replicate the impact of a baseball on a baseball bat [5]. The flywheel had a mechanical stop allowing it to rapidly decelerate and create the illusion of a stronger impact force. Spinshot found that with this model they could generate a force 10 times stronger than previous ungrounded haptic technologies. Rapidly accelerating and decelerating a flywheel to achieve an impact force was a contributing idea to this project.

Although single flywheels are useful for generating impact sensations, they struggle to produce a moment in a single direction. There is an undesirable curving sensation when using a single flywheel.

To solve this problem, a double flywheel system was investigated [6]. A double flywheel system seeks to produce a moment output in a single direction. Studies performed on a dual flywheel model indicate that users can determine the moment direction produced at a rate that is higher than chance. This model was the basis for the dual flywheel system in this project.

2.3 Haptic and Visual Feedback for Learning a New Task

When developing this haptic-virtual reality system, the optimization of user learning was taken into account.

Previous research indicates that haptic guidance can significantly enhance the learning of motor skills by providing immediate, tangible feedback that complements visual and auditory information. Such multisensory integration is particularly effective for acquiring complex spatial and temporal movement patterns, which are crucial in tasks requiring precision and coordination, such as sports simulations like tennis. By replicating the physical sensations associated with racket-ball interactions, haptic systems enable users to experience realistic training scenarios in virtual environments, fostering deeper motor learning through real-time tactile reinforcement [7].

In the context of tennis simulation, haptic feedback allows players to internalize critical aspects of gameplay, including timing, impact force, and racket positioning, by simulating the physical dynamics of racket-ball impacts. For example, systems incorporating directional force feedback create nuanced sensations of impact, which are essential for replicating the realistic feel of a tennis stroke [3]. Likewise, the importance of haptic guidance is highlighted in its ability to reinforce correct movement trajectories and discourage errors, thereby accelerating skill acquisition and retention [7]. These

systems not only replicate real-world conditions but also allow for adaptive feedback based on user performance, making the training experience more personalized and effective.

By bridging the gap between physical and virtual experiences, haptic feedback enables users to refine critical aspects of their gameplay in a controlled, repeatable environment. This synergy between tactile and visual feedback not only enhances immersion but also supports the development of proper stroke mechanics, timing, and consistency. Such systems hold immense potential for advancing skill acquisition in sports, rehabilitation, and other domains requiring precise motor control.

3 DESIGN AND FABRICATION

3.1 The Haptic Racket

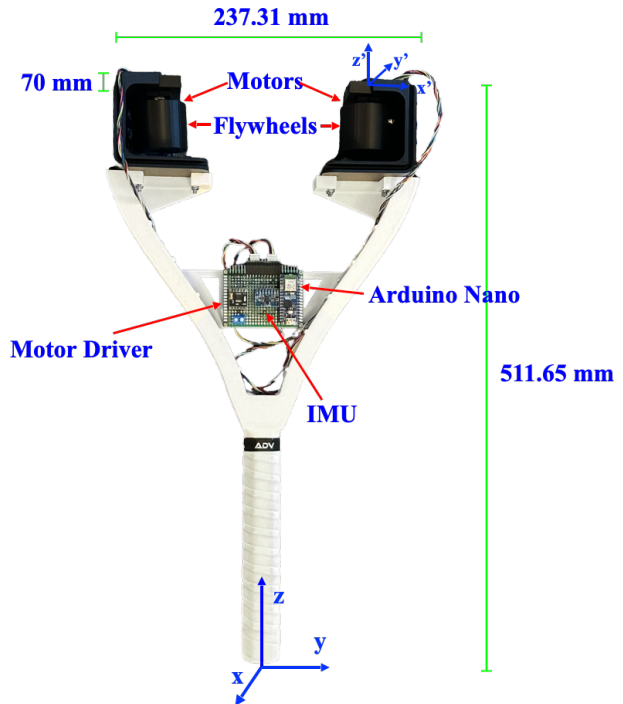


Figure 2: Annotated CAD model of the haptic racket design. Key dimensions are labeled

As shown in Figure 2, the haptic racket was designed with two main goals: to closely replicate the feeling of a real racket, and to properly hold internal components and haptic equipment.

To mimic the feeling of a real racket, overall weight and mass distribution were taken into account. For reference, a standard racket has a mass of 280 - 300 g, and a center of mass around 340 mm from the bottom of the racket. The haptic racket was designed to have a total mass of 333 g and a center of mass 360 mm from the base of the racket. In addition, the ergonomics of the handle were a point of interest as it is what the user is directly interacting with. The handle was modeled to match the grip size of a standard racket. The chosen grip size was 1 (105 mm), selected from the standard tennis racket grip size guide. The handle was also wrapped in a tennis grip to provide additional tactile cues to the user.

The entire structure of the racket was 3D-printed, enabling rapid prototyping and iterative refinements of the design. This fabrication approach allowed for precise integration of the haptic components.

The internal components of this assembly included a protoboard and two 10:1 micro metal gearmotors (Pololu part 5155). The protoboard serves as the central hub for the racket's electronics, housing

an Arduino Nano ESP32 microcontroller, motor driver, and inertial measurement unit (IMU) - an MPU-6050. The protoboard was located on a panel midway between the two arms of the racket (Fig. 2).

The haptic equipment of this assembly included two flywheel-gimbal systems. The flywheel-gimbal assemblies were located on custom platforms at the end of each racket arm (Fig.2). Each flywheel-gimbal assembly consisted of a micro metal gearmotor, a 3D-printed gimbal, and a 3D-printed flywheel. Each flywheel has an outer radius of 48 mm, an inner radius of 18 mm, and a height of 40 mm. The shaft of the motor was secured to the flywheel with a set screw. The gimbal clamped around the body of the motor. The gimbals then attached to the custom platforms on the arms of the racket. The gimbals were designed to be substantially wider than the flywheel's diameter to prevent loose objects from getting caught while the flywheels spin.

3.2 The Virtual Environment

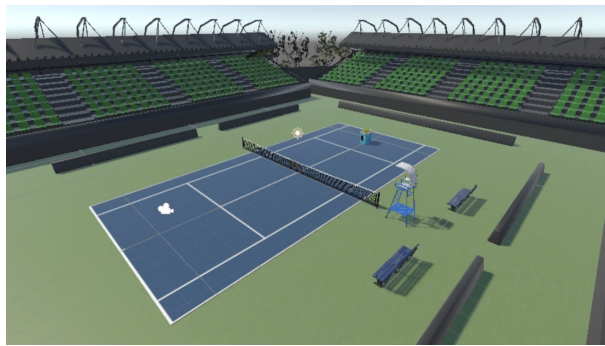


Figure 3: Virtual Environment: Tennis Stadium Simulation

A virtual tennis training environment (Fig. 3) was rendered in Unity. The virtual reality was designed to engage users' visual and audio sensory experiences. The virtual environment allowed users to enter a custom tennis stadium and interact with a virtual racket that was mapped to the haptic racket (Fig. 4). Additionally, audio cues were relayed to the user. The sound of a tennis ball hitting the racket was played through the virtual reality headset each time the ball was struck.

4 METHODS

4.1 Unity Arduino Communications

This section details the integration of Unity I/O, Arduino Nano, and a dual flywheel system for simulating the physical sensations associated with the racket-ball interaction in a tennis simulation. Unity manages the virtual ball simulation, detecting collisions between the racket and ball, while the Arduino Nano controls the dual flywheel system, providing real-time haptic feedback based on the player's interaction with the virtual tennis ball. The system uses the flywheel's inertia and angular velocity to simulate the feeling of ball impact and racket response.

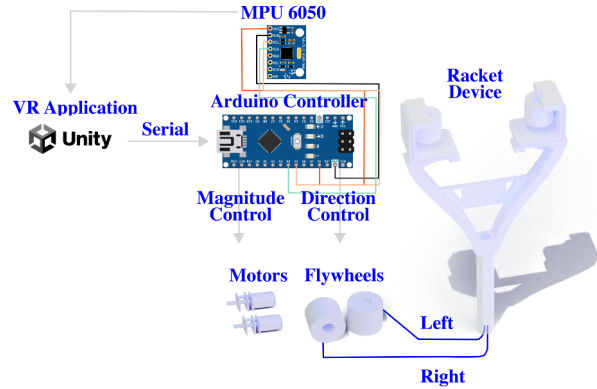


Figure 4: System architecture diagram showing an Arduino controller interfacing with an MPU 6050 IMU for directional and magnitude control. This setup connects to motors and flywheels for generating left and right directional forces on the racket device, integrated with a VR application for immersive interaction.

In the virtual tennis environment, Unity releases tennis balls at 10-second intervals and detects collisions when the IMU sensor on the racket aligns with the ball's position in Unity within a defined threshold. A Kalman filter is applied to the IMU data for improved accuracy, using parameters such as a process noise variance of 0.001 for angle (Q_{angle}), 0.003 for bias (Q_{bias}), and a measurement noise variance of 0.03 ($R_{measure}$). The attractor mechanism ensures consistent ball interaction, minimizing missed collisions. Upon detecting a "hit", Unity sends a collision signal to the Arduino Nano. The Arduino calculates feedback using accelerometer magnitudes, categorized into thresholds (e.g., values over 15,000 trigger a scale factor of 1 and a motor activation duration of 200 ms, while lower values like 6,000 result in a scale factor of 0.7 and 125 ms duration). This feedback is conveyed through a motor-controlled flywheel, emphasizing realistic sensations.

Unity detects the collision between the racket and ball, sending a "hit" signal to the Arduino Nano via a serial communication link. The Arduino Nano receives the "hit" signal from Unity and uses it to activate the dual flywheel system, which provides the physical sensation of ball impact through changes in flywheel angular velocity. The dual flywheel system works by simulating the momentum transfer that occurs when the ball strikes the racket.

In response to the "hit" signal, the Arduino adjusts the flywheel spin time to simulate the intensity and feedback response. When the racket strikes the ball, the dual flywheels spin up, mimicking the momentum and resistance the player would feel in a real-world interaction. Details on how the impact is modeled and rendered are provided in the next section. The feedback intensity depends on the users' swing acceleration, creating a dynamic sensation that simulates the force exerted by the ball. The user's swing acceleration is measured by the IMU mounted on the racket protoboard.

In addition to using haptic feedback to convey ball impact to the user, the racket also features a mode that provides haptic cues to indicate which swing type should be performed. At the beginning of each training session, the user can activate this mode, where only one flywheel is engaged at a time, corresponding to the selected swing type. To determine which swing to focus on, the racket is held with the face flat and parallel to the ground, and the handle positioned perpendicular to the user's body. When the left or right motor spins, it corresponds to a backhand or forehand motion, respectively. This selective activation reinforces the distinction between forehand and backhand motions, allowing users to focus on practicing one swing type at a time and improving their technique within the simulation.

During development, we considered using third-party tools such

as Uduino for integrating Unity with the Arduino Nano. However, we found it more efficient and flexible to write the script ourselves. This approach allowed for greater customization of the communication protocol and system behavior.

4.2 Mathematical Models

To model the impact of the ball on the tennis racket, the mathematical model represented by Equation 1 was used [6]. In Equation 1, the N indicates a Newtonian reference frame and the S/O indicates the moment of S flywheels about the origin (O).

$$\begin{aligned} \vec{N}\vec{M}^{S/O} = & -I_{zz} [\sin \phi (\dot{\omega}_1 - \dot{\omega}_2) + \cos \phi (\omega_1 - \omega_2) \dot{\phi}] \hat{y}' \\ & + I_{zz} [\cos \phi (\dot{\omega}_1 + \dot{\omega}_2) - \sin \phi (\omega_1 + \omega_2) \dot{\phi}] \hat{z} \end{aligned} \quad (1)$$

where:

- ϕ : the rotation of the flywheels spin axis (2)
- ω_1 : the angular velocity of flywheel 1 (3)
- ω_2 : The angular velocity of flywheel 2 (4)
- I_{zz} : The inertia of a singular flywheel (5)

This equation was then simplified. The first of the simplifications was that the two flywheels had equal and opposite accelerations and angular velocities. Additionally, the rotation of the flywheel axis is zero. With these simplifications, the model used in this paper is represented by the equation below [6].

$$M = -2I_{zz}\omega \quad (6)$$

The moment is dependent on the magnitude of the angular velocities of the flywheels. To create the illusion that some moments were stronger than others, the motors were spun for longer periods of time rather than giving them a higher angular velocity. Previous research indicates that a longer duration of impact causes users to believe the force was stronger [2]. This method worked best for this study because the flywheels spun for a short amount of time. The motors did not have enough time to spin up to higher velocities.

The output moment intensity was determined using IMU acceleration data, where the minimum and maximum acceleration magnitudes recorded by the IMU established the range of motion intensities. These acceleration thresholds were divided into smaller intervals and mapped to incrementally increasing motor spin durations in the code, with higher magnitudes (e.g., $a > 15000$) assigned longer durations (200 ms) and lower magnitudes (e.g., $a < 4000$) assigned shorter durations (10 ms) to ensure perceptible differences in haptic feedback (Table 1). Motor spin durations were determined experimentally.

Table 1: Acceleration magnitudes to motor spin time

Acceleration Magnitudes (a)	Motor Spin Time (ms)
$a > 15000$	200
$10000 \leq a < 15000$	175
$6000 \leq a < 10000$	125
$5000 \leq a < 6000$	50
$4000 \leq a < 5000$	25
$a < 4000$	10

5 RESULTS AND DISCUSSION

The haptic racket and virtual reality system were assessed based on performance criteria, including variability in force feedback across different acceleration levels, system responsiveness, and realism. Based on initial informal testing, the force feedback effectively distinguished between varying acceleration levels of the racket, with lower accelerations producing feedback that more closely resembled realistic tennis impacts. At higher accelerations, the flywheels encountered limitations in generating sufficient moments to replicate the sensation of a strong tennis strike accurately. Additionally, the flywheels effectively delivered distinct haptic feedback corresponding to different training modes, such as forehand and backhand cues, demonstrating the potential for skill-specific applications.

A misalignment between visual and haptic cues was observed, resulting in timing inconsistencies. While the haptic feedback was delivered with minimal delay after the virtual ball collision, the discrepancy between the timing of visual and haptic cues highlighted the need for improved synchronization to enhance the system's overall coherence and realism.

These finds indicate that the haptic racket has the potential to provide effective force feedback cues. Enhancement to the flywheel output and improved synchronization of visual and haptic signals would further optimize the system's performance. Replacing the current motors with ones that offer greater torque and higher rotational speed could enable the flywheels to deliver stronger force feedback, particularly at higher accelerations. This improvement would better replicate the sensation of a tennis strike during fast swings, addressing the current limitations in force output.

During the development of the system, challenges were encountered with the MPU-6050 sensor. The primary issues included noisy gyroscopic and accelerometer data, necessitating additional filtering to ensure accurate motion tracking. Notably, the sensor exhibits specific challenges with z-axis measurements. These include significant drift in the z-axis gyroscopic readings, which can lead to inaccuracies over time, and persistent offsets in the z-axis accelerometer data, requiring calibration to ensure accurate performance. These issues necessitated further calibration and algorithmic adjustments to maintain overall system reliability. Furthermore, discrepancies in axis alignment between the IMU and the Unity Game Engine posed a significant problem. The IMU uses a sensor-specific coordinate system, whereas Unity adopts a global coordinate system, leading to misalignment in virtual object movements. Although transformations and calibration were applied to improve synchronization between real-world racket motion and its virtual representation, the Z-axis in the Unity Game Engine remained inaccurate. This inconsistency affected the depth perception of racket movements.

6 CONCLUSION

Future iterations of the system will focus on refining both hardware and software to enhance realism, usability, and reliability. Hardware improvements will include optimizing the racket design by adopting a shorter and sturdier frame, allowing for a reduction in vibrations and better replication of standard tennis racket dimensions and weight distribution. Additionally, a new version of the dual flywheel system would be adopted. This version of the dual flywheel system would also contain servos. The flywheels would rotate continuously and moments would be generated by using the servos to change the orientation of the flywheels [6]. Larger motors with higher torques are also of interest as they would allow for the use of heavier flywheels. Heavier flywheels would allow for the production of stronger forces.

Motion tracking will also be an area of focus. While the current design relies on an IMU, future versions may incorporate a higher-precision IMU with reduced latency or explore Unity's built-in tracking capabilities to enhance synchronization between the physical and virtual components. A Quest controller would be attached to

the haptic racket to rectify the Z-axis misalignment and allow for position tracking.

In terms of software, the attractor mechanism, which was designed to ensure consistent interactions, will be removed in future versions to allow for more naturalistic gameplay. This change will prioritize user skill development by requiring accurate timing and positioning without automated assistance. Enhanced physics simulations in the Unity environment will further improve the realism of ball-racket interactions, with more nuanced collision detection to allow for precise haptic feedback adjustments based on swing dynamics. These planned refinements aim to improve the fidelity of the simulation and broaden its applications in training, rehabilitation, and player engagement.

To build on the insights gained from this development, a future study is planned to systematically evaluate the performance of the haptic racket and virtual reality system. This evaluation will aim to assess the system's ability to deliver realistic force feedback, accurately synchronize visual and haptic cues, and support skill development. To achieve this, future studies should incorporate both quantitative and qualitative psychometric testing to validate the system's effectiveness in enhancing virtual tennis gameplay.

Key hypotheses to investigate include:

- (1) Users will demonstrate improved timing accuracy in ball strikes after training with the haptic feedback system compared to visual-only training (measured through strike timing variance);
- (2) Players will report higher levels of presence and immersion with haptic feedback enabled versus disabled (assessed via presence questionnaires like Igroup Presence Questionnaire (IPQ) or Presence Questionnaire (PQ));
- (3) The haptic feedback will lead to faster motor learning curves for proper stroke mechanics (evaluated through expert assessment of form over multiple training sessions); and
- (4) Players will show better transfer of skills to real tennis play after haptic-enabled versus haptic-disabled virtual training (measured through pre/post real court performance tests).

The force feedback resolution closely replicates realistic tennis impacts at moderate swing speeds, effectively capturing the intensity of motion or impact experienced by the racket. By accurately representing dynamic forces through three-dimensional acceleration measurements, the system provides a reliable platform for meaningful tennis-related studies.² Findings from this evaluation will provide valuable feedback to guide further refinements and validate the system's potential for broader applications in virtual training and skill development.

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²Moderate swing speeds and accelerations are between 6000–15000 magnitude units. Magnitude units are raw sensor outputs measured in least significant bits (LSB) per g, where 1 g (approximately 9.81 m/s²) corresponds to 16384 LSB at the default sensitivity. An acceleration magnitude of 6000–15000 LSB translates to approximately 0.366 g to 0.915 g (3.59 m/s² to 8.97 m/s²).